

docloom: LLM Output to Finished Documents

Two systems. Six formats. Zero compromises.

docloom engine v0.2.0 — 2026-07-21

LLM structured output becomes finished documents

docloom: engine + studio

6

Output formats

14

Block types

993

Tests passing

The engine: validated IR → deterministic renderers

LLM emits JSON. Renderers consume it. Files ship guaranteed.

Render to six formats with one IR

Power-Point

PPTX

native editable

charts

Word

DOCX

hardened tables

Excel

XLSX

formulas & typed cells

Typst

PDF

in-process com- pile

Web Text

HTML & MD

portable export

+

Deterministic. No randomness. Byte-reproducible.

Use 14 block types across three document kinds

DOCX /

PDF /

HTML /

MD

Reports

PPTX

Decks

XLSX

Workbooks

14 total

Block types

8 slide

modes

Layouts

Lint before render, then loop findings back

character & physi- cal

Deck overflow

max 7,

≤130

chars

Slide bullets

max 60

chars

Titles

WCAG

2.x AA

4.5:1

Contrast

every fact

sourced

Citations

Only error-severity findings block render. Warnings surface quality signals for self-correction.

The studio: local-first AI document generation

NotebookLM crossed with Gamma. Runs offline. Fully open.

Generate six artifact kinds from one notebook

PPTX

Decks

**DOCX/
PDF/
HTML/
MD**

Documents

XLSX

Spreadsheets

D2 + SVG

Diagrams

**WAV au-
dio**

Podcast

All offline, fully local-first, no third-party account required.

Grounded retrieval and ranked synthesis

Hybrid dense + lexical retrieval: every answer cites where it came from.

**cosine
over em-
beddings**

Dense similarity

BM25

Lexical rank

**Recipro-
cal Rank
Fusion
(k=60)**

Fusion

**12
chunks**

Evidence per an-
swer

**6 prior
turns**

Chat history

Citations flow into the IR. No hallucinated sources. No unknown citations.

Achieve quality through rigorous testing and safeguards

658

Engine tests

301

Studio tests

34

QA suites

50 MB

Upload limit

**Fernet at
rest**

Encryption

Principles: why we built it this way

Five commitments baked into the architecture:

- Validated IR, not code: LLM output is always a schema, never executable
- Deterministic renderers: same IR + theme = same file, byte-for-byte
- Lint before render: findings loop back, enabling self-correction
- Nothing lost, nothing unreadable: labels never truncate, text always legible
- Local-first, multi-tenant-safe: offline, encrypted, owned sessions